

Problem Set 4

Probability and Statistical Inference

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1. A government is deciding whether to introduce an employment-coaching program. Let β denote the program's effect on the probability of employment, measured in percentage points. A positive value of β means that the program raises the employment probability. The government will roll out the program only if there is evidence that it improves employment.
 - (a) Write down suitable null and alternative hypotheses for this pre-specified question about whether the program improves employment.
 - (b) Describe a Type I error and a Type II error in terms of the program and the rollout decision.
 - (c) Suppose rolling out an ineffective program would be especially costly. Holding the design and sample size fixed, explain why the government might choose a smaller significance level, and state the resulting trade-off.
 - (d) The study produces a one-sided p -value of 0.08. What decision is made at the 5% level? Explain what the value 0.08 does and does not mean.
 - (e) In a separate scenario, an analyst sees a negative estimate and proposes changing the pre-specified right-sided test to a left-sided test. Explain why this would not be a valid way to conduct the pre-specified test. How could concern about effects in either direction have been handled?

2. An estimated regression coefficient is $\hat{\beta} = 0.18$ with estimated standard error 0.07. Consider $H_0 : \beta = 0.05$. There are 76 degrees of freedom, and you may use

$$F_{76}(1.857) = 0.9663, \quad t_{0.95,76} = 1.665, \quad t_{0.975,76} = 1.992,$$

where F_{76} is the cumulative distribution function (cdf) of a t_{76} random variable.

- (a) Calculate the test statistic.
 - (b) Test against $H_1 : \beta > 0.05$ at the 5% level in three equivalent ways: (i) using a critical value, (ii) using a p -value, and (iii) **non-examinable extension:** using the corresponding one-sided confidence set. Verify that the methods give the same decision.
 - (c) For $H_1 : \beta \neq 0.05$, carry out a two-sided 5% test using a critical value, a p -value and a 95% confidence interval. Verify that all three decisions agree.
 - (d) Explain why the two alternatives can produce different decisions from the same estimate. Under which regression assumptions and standard-error calculation is the t_{76} reference distribution exact?
3. **Non-examinable extension: inverting tests.** To invert a test means to find all null values that the test would not reject. Let $c > 0$ be the appropriate critical value, and treat $\widehat{se}(\hat{\beta})$ as fixed during this calculation.

- (a) Starting from the non-rejection condition

$$\left| \frac{\hat{\beta} - \beta_0}{\widehat{se}(\hat{\beta})} \right| \leq c,$$

solve for all null values β_0 that would not be rejected. Hence derive the usual two-sided confidence interval.

- (b) For tests against $H_1 : \beta > \beta_0$, non-rejection occurs when $(\hat{\beta} - \beta_0)/\widehat{se}(\hat{\beta}) \leq c$. Invert this inequality to derive the corresponding one-sided confidence set.
 - (c) Use your answers to explain the statement: “a confidence set is the collection of null values not rejected by the corresponding tests.”
4. Let $Y_1, \dots, Y_n$ be independent $N(\mu, 12^2)$ observations. We test $H_0 : \mu = 100$. Suppose first that $n = 36$, so the standard error of $\bar{Y}$ is 2, and consider the alternative value $\mu = 104$. Power at $\mu = 104$ means the probability of rejecting the null when the true mean is 104. Use

$$z_{0.95} = 1.645, \quad z_{0.975} = 1.960, \quad z_{0.80} = 0.842,$$

as well as $\Phi(0.355) = 0.6387$, $\Phi(0.04) = 0.5160$, and $\Phi(-3.96) \simeq 0.00004$, and $\Phi(-3.645) \simeq 0.00013$.

- (a) Derive the rejection rule and power at $\mu = 104$ for a right-sided 5% test.
 - (b) Derive the rejection rule and power at $\mu = 104$ for a two-sided 5% test.
 - (c) Explain the source of the difference between the two powers. What is the approximate power of the right-sided test if the true mean is instead 96?
 - (d) Find the smallest integer n giving at least 80% power at $\mu = 104$ for the right-sided test.
 - (e) Using the usual minimum-detectable-effect (MDE) approximation, find the required n for 80% power at $\mu = 104$ with a two-sided test. What happens to the standard error and the MDE if n is quadrupled?
5. A research team conducts $m = 20$ independent tests of different outcomes and reports the outcome that gives the strongest apparent evidence against its null hypothesis (the smallest p -value). Assume all 20 null hypotheses are true. For the exact probability calculations, assume each test's actual rejection probability equals its chosen significance level: 0.05 initially and 0.05/20 after the Bonferroni correction.
- (a) Find the probability of at least one false rejection, which is the family-wise error rate. Distinguish this probability from the expected number of false rejections.
 - (b) Under a Bonferroni correction designed to control the family-wise error rate at 0.05, what level is used for each test?
 - (c) Use the union bound to show that Bonferroni controls the family-wise error rate even without independence. Under independence, calculate its exact family-wise error rate.
 - (d) The smallest unadjusted p -value is 0.018. Is it significant under the Bonferroni rule? Explain how pre-specifying a primary outcome changes the multiple-testing problem.
 - (e) In a separate Monte Carlo exercise (a repeated computer simulation) with $R = 100,000$ repetitions, a nominal 5% right-sided test rejects 5,041 times under the null and 63,870 times at a specified alternative. Estimate size and power. Explain the different effects of increasing the number of simulation repetitions R and increasing the study sample size n .